

IEOR E4004

Optimization Models and Methods

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Outline

- Syllabus Review
- Introduction to IEOR 4004 Optimization Models and Methods
- Who Uses Optimization?
- Burrito Optimization Game
- Introduction to Mathematical Modeling

IEORE4004

Syllabus Review

Link to course syllabus:

https://courseworks2.columbia.edu/courses/238858/files/26391664?module_item_id=3448462



COLUMBIA ENGINEERING

The Fu Foundation School
of Engineering and Applied Science



Introduction to Operations Research and Optimization Models and Methods

Topics

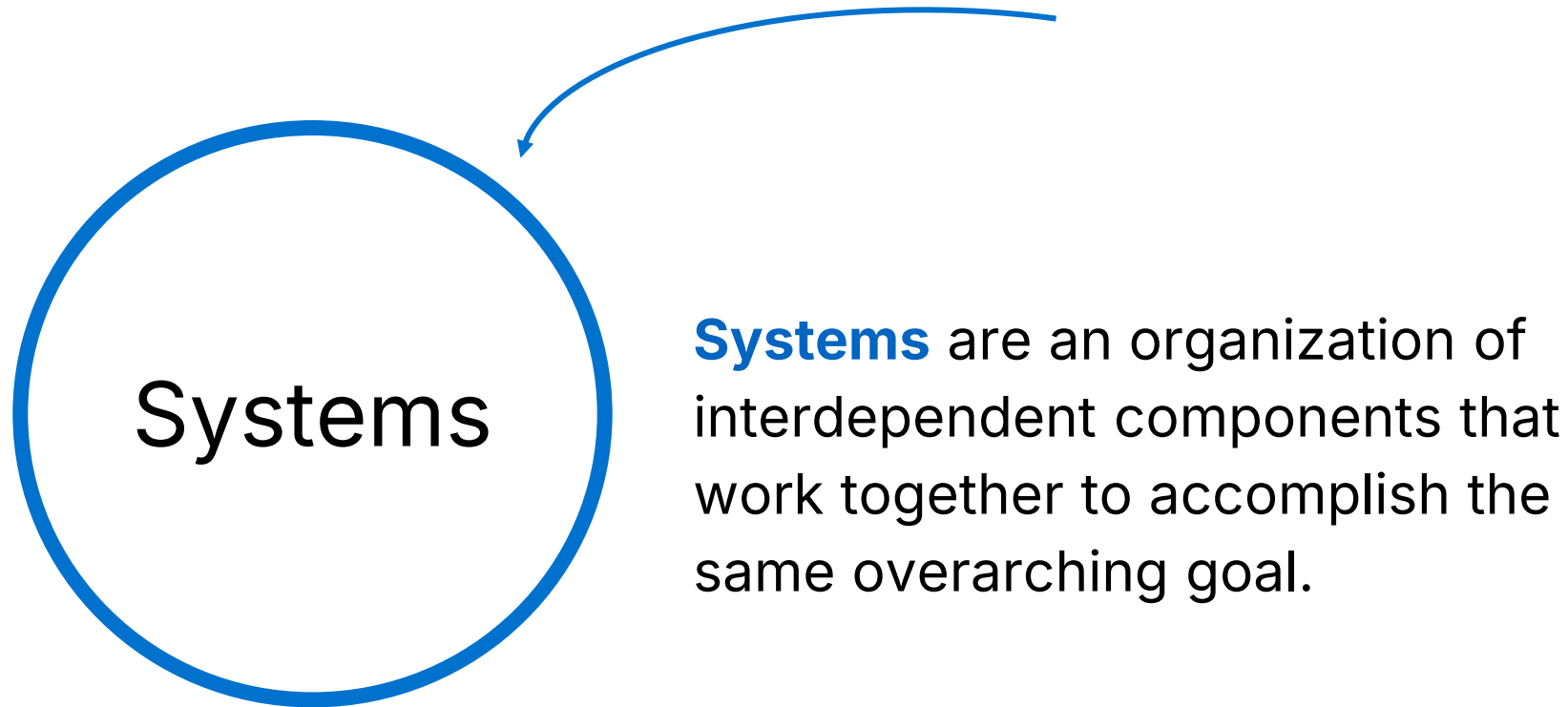
- What is operations research (OR)?
- What is mathematical modeling?
- What is optimization?
- Who uses it?

What is Operations Research (OR)?

Operations research (often referred to as **management science**) is a **scientific approach** to decision making that seeks to best design and operate a **system**

What is OR?

Operations research is a **scientific approach** to decision making that seeks to best design and operate a **system**



What is OR?

Some examples of **Systems**:

- Transportation systems



What is OR?

Some examples of **Systems**:

- Transportation systems
- Telephone systems



What is OR?

Some examples of **Systems**:

- Transportation systems
- Telephone systems
- Healthcare systems



What is OR?

Some examples of **Systems**:

- Transportation systems
- Telephone systems
- Healthcare systems
- Ecological systems
- ...



What is OR?

Operations research is a **scientific approach** to decision making that seeks to best design and operate a **system**



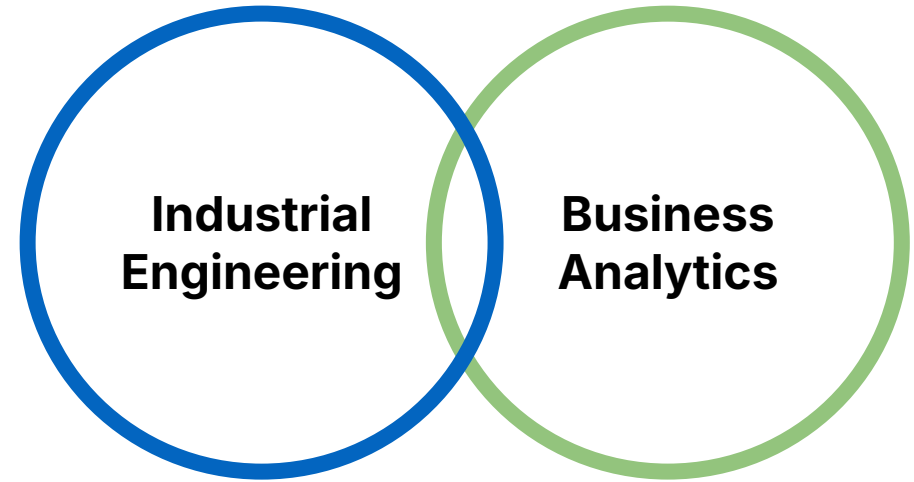
Scientific
approach

The **scientific approach** to decision making usually involves the use of one or more mathematical models:

- Prediction models
- Simulation models
- Network models
- Optimization models...

What is OR?

Similar techniques are used in **industrial engineering** with a somewhat narrower view of applications.



There is a strong overlap with **business analytics**, which refers to any use of data to improve business processes, whether statistical methods or optimization.

The Origins of OR

- Origins during **World War II**

- Scientists worked on military problems on request of military leaders

- ex: deployment of radars, management of anti submarine operations

- Post-war shifts to **broader applications**

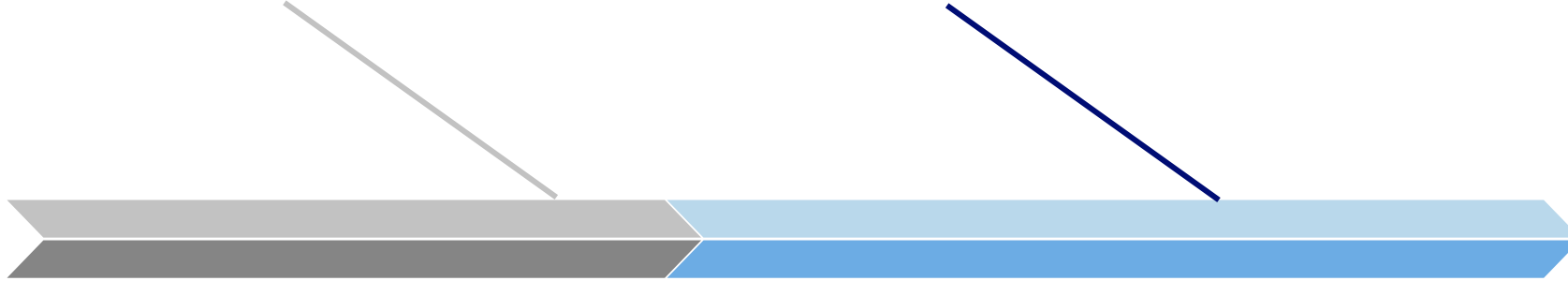
- Shift from military to governmental, industrial, and business applications

- ex: logistics, supply chain management, resource allocation

The Origins of OR

World War II: work on military problems

Post-War: focus on governmental, industrial and business applications



The field had its **origins** during World War II.

Models used in these applications tend to have many **interrelated variables, significant amounts of data requirement, and complex structure** unique to the application.

Who Uses Optimization?

Businesses



Governmental Organizations



Nonprofit Organizations



Who Uses Optimization?

In addition to data and statistical analysis, **a mathematical model of the system** is often needed to find the best or even a viable option.

Who Uses Optimization?

Example decisions include:

- Planning the flow of goods through a supply chain



Who Uses Optimization?

Example decisions include:

- Planning the flow of goods through a supply chain
- Scheduling personnel shifts at a hospital



Who Uses Optimization?

Example decisions include:

- Planning the flow of goods through a supply chain
- Scheduling personnel shifts at a hospital
- Choosing an investment portfolio



What Is Optimization?

The approach of optimization modeling involves developing a mathematical model by:

1. Describing which potential **decisions** could be taken,
2. Considering physical, logical, financial, or other **limitations**, and
3. Assessing them using a single **objective**

The mathematical model developed in this structure is an **optimization problem**; the study of them is called **optimization**

Research Areas | About the Instructor

→ Improving Housing Access for Unhoused Youth in New York City

1. Assessing the required increase in resources to provide appropriate housing and support services for New York City's unhoused youth



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1. Assessing the required increase in resources to provide appropriate housing and support services for New York City's unhoused youth
2. Evaluating policies to reduce housing access disparities among different demographic groups



Research Areas | About the Instructor

→ Improving Housing Access for Unhoused Youth in New York City

1. Assessing the required increase in resources to provide appropriate housing and support services for New York City's unhoused youth
2. Evaluating policies to reduce housing access disparities among different demographic groups
3. Working with community stakeholders to improve current public housing system's technological infrastructure



Research Areas | About the Instructor

→ Healthcare Scheduling Optimization Models for Improved Patient Access

Developing efficient scheduling strategies for:

- Improved radiology access by ensuring that at least 90% of emergency department patients undergo MRI scans within one hour of their arrival ([radiology scheduling](#))



Research Areas | About the Instructor

→ Healthcare Scheduling Optimization Models for Improved Patient Access

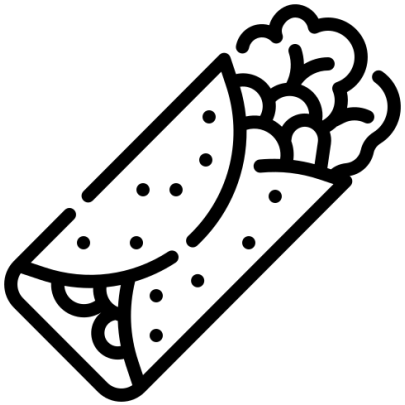
Developing efficient scheduling strategies for:

- Improved radiology access by ensuring that at least 90% of emergency department patients undergo MRI scans within one hour of their arrival ([radiology scheduling](#))
- Improved lung cancer screening by prioritizing at risk populations ([oncology screening](#))



Burrito Optimization Game

A Facility Location Problem

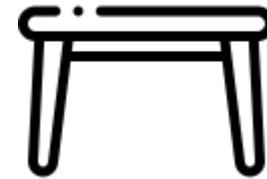
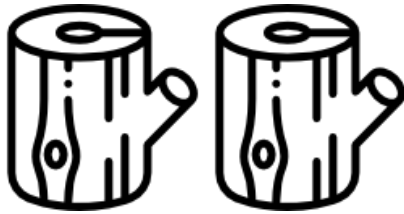


Introduction to Mathematical Modeling

A Production Problem

Mathematical Modeling

Consider a company produces chairs and tables:



Mathematical Modeling



Consider a company produces chairs and tables:

	Table	Chair
Wood need per unit	2	1
Hours of labor needs per unit	1	1
Profit of selling one item	\$3	\$2
Limit of selling each week	40	No limit

Each week, we can have **80 hrs of labor** and **100 units of wood** to spend.

Let us find out, what is the weekly production plan that maximize the profit?

A Production Problem: Formulation



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1. Describing which potential **decisions** could be taken,
2. Considering physical, logical, financial, or other **limitations**, and
3. Assessing them using a single **objective**

A Production Problem: Formulation



Guiding Questions:

What are the **potential decisions** that we can take?

What are our **constraints/ limitations**?

What is our **objective** in this problem?

A Production Problem: Formulation



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Step 1: Define decision variables

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Decision Variables

A Production Problem: Formulation



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Step 2: Form objective function

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Objective

A Production Problem: Formulation



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Step 2: Form objective function

Step 3: Add Constraints

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A Production Problem: Formulation



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Constraints

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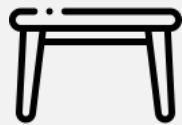
Let us find out, what is the weekly production plan that maximize the profit?

A Production Problem: Formulation



Step 1: Define decision variables

01



= Number of tables we should produce in a week



= Number of chairs we should produce in a week

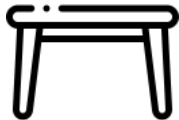
A Production Problem: Formulation



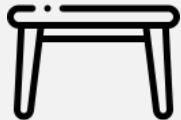
Step 2: Form objective functions

Q: What is our aim/objective?

→ To find the production plan that will **maximize** the profit

Profit: 3  + 2 

02

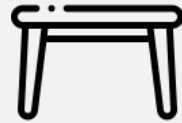
Aim/objective: **maximize** 3  + 2 

A Production Problem: Formulation



Step 3: Add constraints

Remember:



produced each week = x_1



produced each week = x_2

A Production Problem: Formulation



Step 3: Add constraints

03

subject to: 2  +  ≤ 100 Capacity of wood

A Production Problem: Formulation



Step 3: Add constraints

subject to $2 \begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} + \begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} \leq 100$ Capacity of wood

$\begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} + \begin{array}{c} \text{---} \\ | \quad | \\ \text{---} \end{array} \leq 80$ Upper limit on labor

A Production Problem: Formulation



Step 3: Add constraints

$$\text{subject to } 2 \text{  + \text{  } \leq 100$$

Capacity of wood

$$\text{ + \text{  } \leq 80$$

Upper limit on labor

$$\text{ } \leq 40$$

Demand for tables

A Production Problem: Formulation



Step 3: Add constraints

$$\text{subject to } 2 \text{  + \text{  \leq 100$$

Capacity of wood

$$\text{ + \text{  \leq 80$$

Upper limit on labor

$$\text{ \leq 40$$

Demand for tables

$$\text{ , \text{  \geq 0$$

Non-negativity

A Production Problem: Algebraic Formulation



Step 1: Define decision variables

Step 2: Form objective function

Step 3: Add Constraints

01

x_1 : number of tables to produce in a week
 x_2 : number of chairs to produce in a week

02

$\text{maximize}_{x_1, x_2} 3x_1 + 2x_2$ Profit

03

subject to $2x_1 + x_2 \leq 100$ Capacity of wood
 $x_1 + x_2 \leq 80$ Upper limit on labor
 $x_1 \leq 40$ Demand for tables
 $x_1, x_2 \geq 0$ Non-negativity

A Production Problem: Algebraic Formulation

Step 1:

Define decision variables

x_1 : number of
tables to produce
in a week

x_2 : number of
chairs to produce
in a week

Step 2:

Form objective function

maximize x_1, x_2
profit: $3x_1 + 2x_2$

Step 3:

Add constraints

Capacity of woods:
subject to $2x_1 + x_2 \leq 100$

Upper limit on labor
 $x_1 + x_2 \leq 80$

Demand for tables
 $x_1 \leq 40$

Non-negativity
 $x_1, x_2 \geq 0$